INCH-POUND

MIL-C-70714 (AR)
AMENDMENT 6
26 December 1992
SUPERSEDING
AMENDMENT 5
25 September 1992

MILITARY SPECIFICATION

CARTRIDGE, 60MM, ILLUMINATING, M721
LOADING, ASSEMBLING, AND PACKING

This amendment forms a part of MIL-C-70714 (AR), dated 28 August 1987, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Add "(M721)" after "XM721"; and wherever XM721 appears in the specification.

PAGE 2

2.1.2, Government drawings: PRODUCT AND PACKAGING DRAWINGS: Change "XM776" to "M776"; and wherever XM776 appears in the specification.

Change title of 9354466 as follows:
"Container, Ammunition, Metal, PA124".

Change title of 9390711 as follows:

"Packing and Marking for Box, Wirebound, for Cartridge, 60MM, Illuminating, M721".

Inspection Equipment...

INSPECTION EQUIPMENT DRAWINGS: Delete "7297287 - Chamber Gage" and "7257965 - Limit Check Gage" and add "7256348 - Alignment Gage."

PAGE 3

3.2.1, Add new paragraph as follows:

"3.2.1 Tail Cone. The inside surface of the tail cone will be free of dimples after drilling holes and staking of the shear pin. A dimple is considered to be a raised surface exceeding 0.010 inches high."

AMSC N/A FSC 1310

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PAGE 4

3.5.6, fuze times: Change, "19.0" to "21.0" (2 places) and "26.0" to "27.0" (2 places).

PAGE 6

Table II, Examination or test: Change "Fast" to "First" and change under cartridge examination, the sample size (2 places) from "50" to "176."

PAGE 8

4.4.2.1, Major 107: Delete in its entirety.

Add new major 108 as follows:

"108 Compression pad(s) missing or improperly placed...0.40%... 3.2...Visual."

PAGE 9

4.4.2.2, Major 103: Delete in its entirety.

PAGE 10

4.4.2.3, Major 106: Change "Visual" to "Gage".

Delete major 101 and 102 in their entirety.

Add new majors 108 and 109 as follows:

"108 Tail cone improperly assembled to the body tube assembly ...(a)...3.2.1...Visual/gage.

109 Presence of dimple on inside surface of tail cone...(a)... 3.2.1...Visual/gage."

Change note (a) to read as follows:

"(a) Defect callout and alternate sampling plan are depicted in 4.4.3.4"

PAGE 12

4.4.2.5, Major 101: Delete inspection method "Gage" and substitute "7256348."

Major 108: Delete inspection method "Visual" and substitute "Gage."

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Major 109 and 110 and Note 1: Delete in their entirety.

PAGE 14

4.4.2.6, Add Major 103 as follows:

"103, Inner Tube missing ... 0.40% ... 5.1 ... Visual."

PAGE 15

* 4.4.2.7, Under Minor 201, change "5.1" to "5.3"

PAGE 16

* 4.4.2.8, Title: Change "Contained" to "Container."

Delete Major 104 in its entirety.

Add new Minor as follows:

"201 Graphical firing scale missing ...0.65% ... 5.2 ... Visual".

PAGE 17

* 4.4.2.9, Title: Change "Contained" to "Container."

Under Major 101, change "5.2" to "5.3"

PAGE 18

4.4.3.2.1.1, Change "85" to "200."

4.4.3.3, Last sentence: Delete end of sentence as follows:

"made on the applicable...the rejected production."

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4.4.3.4, Delete in its entirety and substitute the following:

"4.4.3.4 Alternate sampling plan for examination of depth, diameter of shear pin holes and presence of dimples. The depth and diameter of the shear pin holes and presence of dimples shall be inspected in the following manner using approved gages. Beginning with the first production shift, ten (10) consecutive drilled inert units shall be inspected and found defect free prior to rate production. Then inspect five (5) units every two (2) hours until the end of the shift. Five (5) units shall be inspected at the end of the shift. If a defect is found in any of the five (5) units, the machine set up shall be adjusted until ten (10) consecutive units are defect free.

"4.4.3.4.1 Tail cone and body tube inspection. The projectile loading assemblies shall be subjected to a load (pull) to shear off the pins; slowly remove the body tube assembly from the tail cone and once the canister is visible, hold the canister down to prevent movement from the tail cone while removing the body tube. Inspect the cable by lifting candle slightly, if one coil of the cable is above the cable protectors, the sample shall be classed defective. Release the parachute expulsion spring; if the parachute folding and packing assembly does not pop out freely, the sample shall be classed as defective. If the drogue chute is tangled or improperly packed, the sample shall be classed as defective. If the cable is frayed, kinked or deformed, the sample shall be classed as defective. Inspect the inside surface of the tail cone where the shear pin holes were drilled for "dimpling." If the presence of two dimples 180 degrees apart are shown, the sample shall be classed as defective.

4.4.3.4.2 Production shift examination. Beginning with the first production shift ten (10) consecutive projectile assemblies shall be inspected and found defect free prior to rate production. Then inspect five (5) units randomly selected for every two (2) hours until the end of the shift. Five (5) units shall be inspected at the end of the shift. If a defective unit is found in any group of five (5), the production represented by the units shall be rejected and the machine set up shall be adjusted until ten (10) consecutive units are defect free.

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PAGE 21 (Cont'd)

4.4.5.7, Line 19: Delete sentence starting with "the cartridges shall be conditioned..."

Test settings: Change "19.0" to "21.0" and "26.0" to "27.0."

Add to paragraph as follows:

"Meteorological data should be checked prior to and during testing. If the wind conditions exceed the TECOM firing table TOP thresholds or following criteria, then the test should be halted until the conditions are favorable.

a. When firing to maximum ordinates below 1,500 meters:

1. Surface winds exceed 10 knots.

2. Gust winds exceed 5 knots.

b. When firing to maximum ordinates above 1,500 meters.

1. Surface winds exceed 20 knots.

2. If the wind speed changes by more than 15 knots or the wind direction changes by 800 mils when compared to two successive rawinsondes at a constant height. If the directional change exceeds 800 mils but the wind speed is less than 5 knots at the comparison height, firing may continue."

PAGE 25

4.4.5.8, Delete in its entirety.

The margins of this amendment are marked with an asterisk, or vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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4.4.3.4.3 Alternate sampling plan procedure. The alternate sampling plan procedure depicted in 4.4.3.4.2 shall be applicable to each drilling machine/equipment setup.

4.4.3.4.4. Tooling change. A change of tooling will require a first piece inspection prior to continuing production. The sampling plan will be the same as that prior to the tooling change.

4.4.3.5, Delete first sentence in its entirety and substitute the following:

"One hundred and seventy-six (176) sample cartridges shall be selected for first article testing and fifty sample cartridges shall be selected for production lot testing. The cartridges for first article testing shall be separated into three groups; one group (48 samples) conditioned for 24 hours at plus 145 degrees F $\pm$ 2.5 degrees F, one group (80 samples) conditoned for 24 hours at plus 70 degrees F $\pm$ 2.5 degrees F, and one group (48 samples) conditioned for 24 hours at minus 50 degrees F $\pm$ 2.5 degrees F. Each group shall be separated into equal subgroups; one subgroup tested at charge zone 1 and one subgroup tested at charge zone 4. The cartridges for production lot testing shall be conditioned for 24 hours at plus 70 degrees F $\pm$ 2.5 degrees F."

4.4.3.5.1, Delete paragraph in its entirety and substitute the following:

"4.4.3.5.1 Overall acceptance criteria. In addition to the individual accept/reject criteria of 4.4.3.5, the first article sample shall be rejected in a total of five or more major defects occur. The production lot sample shall be rejected if a total of three or more major defects occur."

4.4.3.6, Delete in its entirety.

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